

Numerical Interpretation Interpolation

Newton's forward and backward difference interpolation formula.

Operators : $\Delta, \nabla, u, \delta, E, D$

Interpolation

To find intermediate value in an interval of a function is known as interpolation.

Extrapolation

To find value of a function outside the given interval is known as extrapolation.

eg: Consider a set of data points for a function as

x	y
2	3
3	4
4	7
5	6
6	8

Suppose we want to find the values at the points $x=2.2, 4.1, 5.9$ then the process is known as interpolation.

Suppose we want to find the value at $x=6.2$, then the process is known as extrapolation.

10

12

1

2

3

4

5

6

2

The first order
corresponding to the
given data

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points $x: x_0, x_1, x_2, x_3, x_4$
 $y: y_0, y_1, y_2, y_3, y_4$

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of forward difference. first order forward

Clearly, we define the first order forward differences as: $y_1 - y_0 = \Delta y_0$

$$y_2 - y_1 = \Delta y_1$$

$$y_3 - y_2 = \Delta y_2$$

$$y_{n+1} - y_n = \Delta y_n$$

The higher ordered differences are defined by

$$\Delta y_1 - \Delta y_0 = \Delta^2 y_0$$

$$\Delta y_2 - \Delta y_1 = \Delta^2 y_1$$

Similarly, the third order differences can be defined by

$$\Delta^2 y_1 - \Delta^2 y_0 = \Delta^3 y_0$$

Forward difference table for the given data points

<u>x</u>	<u>y</u>	<u>Δy</u>	<u>$\Delta^2 y$</u>	<u>$\Delta^3 y$</u>	<u>$\Delta^4 y$</u>
2	3				
3	4	1			
4	5	1	0		
5	8	3	2	2	
6	2	-6	-9	-11	-13

9

11

3

5

6

7

$$\Delta^2 f(x) = \Delta(\Delta f(x))$$

$$= \Delta(f(x+h) - f(x))$$

$$\neq \cancel{f(x+h) - f(x+h)} = \cancel{f(x+h) + f(x)}$$

$$= \Delta f(x+h) - \Delta f(x)$$

$$= f(x+h+h) - f(x+h) - (f(x+h) - f(x))$$

$$= f(x+2h) - 2f(x+h) + f(x)$$

General formula for backward difference

Backward difference table

Values are used to interpolate at the end of an interval.

The first order backward differences corresponding to the data points

$$x: x_0, x_1, x_2, x_3, x_4$$

$$y: y_0, y_1, y_2, y_3, y_4$$

of a function $y = f(x)$ is defined as

$$y_1 - y_0 = \nabla y_1$$

$$y_2 - y_1 = \nabla y_2$$

$$y_3 - y_2 = \nabla y_3$$

11/14 higher order differences are defined as
 $\nabla y_2 - \nabla y_1 = \nabla^2 y_2$ etc.

Backward difference table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_0	y_0				
x_1	y_1	∇y_1			
x_2	y_2	∇y_2	$\nabla^2 y_2$		
x_3	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_3$	
x_4	y_4	∇y_4	$\nabla^2 y_4$	$\nabla^3 y_4$	$\nabla^4 y_4$

Consider the data points

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
2	3				
3	4	+1			
4	7	3	2		
5	6	-1	-4	-6	
6	8	2	3	7	13

$$y_4 = 8$$

$$\nabla y_4 = 2$$

$$\nabla^2 y_4 = 3$$

$$\nabla^3 y_4 = 7$$

$$\nabla^4 y_4 = 13$$

$$\nabla f(x) = f(x) - f(x-h)$$

$$\nabla f(x) = f(x) - f(x-h)$$